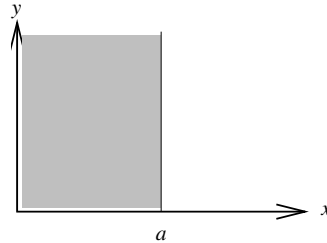


## PDEs

## Solutions to Problem Sheet 6

1. (a) The region is sketched below.



$$V(x, y) = Ae^{-n\pi y/a} \sin\left(\frac{n\pi x}{a}\right).$$

Then

$$\begin{aligned}\frac{\partial^2 V}{\partial x^2} &= -\frac{n^2\pi^2}{a^2} Ae^{-n\pi y/a} \sin\left(\frac{n\pi x}{a}\right), \\ \frac{\partial^2 V}{\partial y^2} &= \frac{n^2\pi^2}{a^2} Ae^{-n\pi y/a} \sin\left(\frac{n\pi x}{a}\right).\end{aligned}$$

Hence,  $V_{xx} + V_{yy} = 0$ .

Since  $e^{-n\pi y/a} \rightarrow 0$  as  $y \rightarrow \infty$ ,  $V \rightarrow 0$  as  $y \rightarrow \infty$ .

$V(0, y) = 0$  and  $V(a, y) = 0$  since  $\sin(0) = 0$  and  $\sin(n\pi) = 0$  for  $n$  integer.

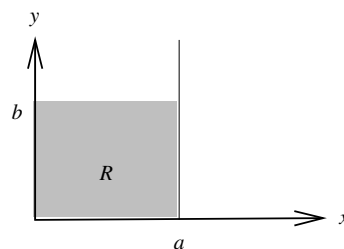
$V(x, 0) = A \sin(n\pi x/a) = 100 \sin(2\pi x/a)$  when  $A = 100$  and  $n = 2$ .

Hence,

$$V(x, y) = 100e^{-2\pi y/a} \sin(2\pi x/a),$$

satisfies Laplace's equation in  $R$ , along with the given boundary conditions.

- (b) The region is sketched below



$$V(x, y) = A + B \sinh\left(\frac{n\pi y}{a}\right) \cos\left(\frac{n\pi x}{a}\right).$$

Then

$$\begin{aligned}\frac{\partial V}{\partial x} &= -\frac{n\pi}{a}B \sinh\left(\frac{n\pi y}{a}\right) \sin\left(\frac{n\pi x}{a}\right), \\ \frac{\partial^2 V}{\partial x^2} &= -\frac{n^2\pi^2}{a^2}B \sinh\left(\frac{n\pi y}{a}\right) \cos\left(\frac{n\pi x}{a}\right), \\ \frac{\partial^2 V}{\partial y^2} &= \frac{n^2\pi^2}{a^2}B \sinh\left(\frac{n\pi y}{a}\right) \cos\left(\frac{n\pi x}{a}\right).\end{aligned}$$

Hence,  $V_{xx} + V_{yy} = 0$ .

$\frac{\partial V}{\partial x} = 0$  when  $x = 0$  and  $x = a$  since  $\sin(0)$  and  $\sin(n\pi) = 0$  for  $n$  integer.

$V(x, 0) = A$  since  $\sinh(0) = 0$ , hence  $V(x, 0) = 0$  when  $A = 0$ .

Let  $y = b$ . Then

$$V(x, b) = B \sinh\left(\frac{n\pi b}{a}\right) \cos\left(\frac{n\pi x}{a}\right) = \cos\left(\frac{\pi x}{a}\right)$$

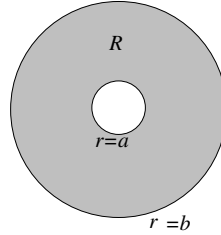
when  $n = 1$  and  $B = 1/\sinh\left(\frac{\pi b}{a}\right)$ .

Hence,

$$V(x, y) = \frac{\sinh\left(\frac{\pi y}{a}\right) \cos\left(\frac{\pi x}{a}\right)}{\sinh\left(\frac{\pi b}{a}\right)}$$

satisfies Laplace's equation in  $R$ , along with the given boundary conditions.

2. The region is sketched below



$$T(r, \theta) = A + B \ln r, \quad \text{for } r > 0.$$

Then

$$\frac{\partial T}{\partial r} = \frac{B}{r}, \quad \frac{\partial^2 T}{\partial r^2} = -\frac{B}{r^2}, \quad \frac{\partial^2 T}{\partial \theta^2} = 0.$$

Hence,  $T_{rr} + \frac{1}{r}T_r + \frac{1}{r^2}T_{\theta\theta} = 0$ .

$T(a, \theta) = T_1$  and  $T(b, \theta) = T_2$ . Hence,

$$A + B \ln a = T_1, \quad A + B \ln b = T_2.$$

Solving these equations for  $A$  and  $B$  we obtain

$$B = \frac{T_1 - T_2}{\ln a - \ln b} = \frac{T_1 - T_2}{\ln(a/b)},$$

and

$$A = \frac{1}{2}(T_1 + T_2) - \frac{1}{2}(T_1 - T_2) \frac{\ln(ab)}{\ln(a/b)}.$$

Hence,

$$T(r, \theta) = \frac{1}{2}(T_1 + T_2) - \frac{1}{2}(T_1 - T_2) \frac{\ln(ab)}{\ln(a/b)} + \frac{(T_1 - T_2)}{\ln(a/b)} \ln r$$

satisfies Laplace's equation, along with the given boundary conditions.

3.

$$U(x, t) = A \sin\left(\frac{n\pi x}{a}\right) \exp\left(-\frac{n^2\pi^2 K t}{a^2}\right).$$

Then

$$U_{xx} = -\frac{n^2\pi^2}{a^2} A \sin\left(\frac{n\pi x}{a}\right) \exp\left(-\frac{n^2\pi^2 K t}{a^2}\right),$$

$$U_t = -\frac{n^2\pi^2 K}{a^2} A \sin\left(\frac{n\pi x}{a}\right) \exp\left(-\frac{n^2\pi^2 K t}{a^2}\right).$$

Hence,  $U_{xx} - \frac{1}{K} U_t = 0$ .

$U(0, t) = 0$  and  $U(a, t) = 0$  since  $\sin(0) = 0$  and  $\sin(n\pi) = 0$  for  $n$  integer.

$U(x, 0) = A \sin\left(\frac{n\pi x}{a}\right) = 10 \sin\left(\frac{\pi x}{a}\right)$  when  $n = 1$  and  $A = 10$ .

Hence,

$$U(x, t) = 10 \sin\left(\frac{\pi x}{a}\right) \exp\left(-\frac{\pi^2 K t}{a^2}\right)$$

satisfies the heat equation in the bar, along with the given boundary and initial conditions.